

Vilnius University Šiauliai Academy



Object Oriented Programming
Practical Work 2:

OOP: Object, Methods, Access Modifiers, \$this, Constructor

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Test Report

Tasks:

1. Rewrite $3x+1$ task to OOP.
2. The class part move to separate file.
3. Append Constructor to add start Nummber.
4. Create a separate method, perform calculations based on a given interval (e.g.: 25-24658)
5. Create a method *statistics* that returns: the number with max iterations, the number with min iterations, the number with max reached value.

Test 1: Program initialization

```
C:\Users\Acer\Desktop\3XPLUS100P_Assignment2>php C:\Users\Acer\Desktop\3XPLUS100P_Assignment2\3xplusloop.php
--Welcome, here is a practical demonstration of the Collatz sequence!--
Enter a number to receive Collatz (number must be >= 1):
```

Result: The program successfully initializes upon running the main file (3xplusloop.php).

Test 2: Input validation of letters, special characters, and zero

```
--Welcome, here is a practical demonstration of the Collatz sequence!--
Enter a number to receive Collatz (number must be >= 1): -10
Enter. a number. above. or equal. to 1.
Enter a number to receive Collatz (number must be >= 1): a
Enter. a number. above. or equal. to 1.
Enter a number to receive Collatz (number must be >= 1): *
Enter. a number. above. or equal. to 1.
Enter a number to receive Collatz (number must be >= 1): 0
Enter. a number. above. or equal. to 1.
Enter a number to receive Collatz (number must be >= 1):
```

Result: The program successfully prohibits the user from running the program, and reverts them back to the previous input state.

Test 3: Correct input test and calculation accuracy

```
--Welcome, here is a practical demonstration of the Collatz sequence!--
Enter a number to receive Collatz (number must be >= 1): 5
Thank you. Here is the Collatz sequence!
-----
Sequence step 1: 5
Sequence step 2: 16
Sequence step 3: 8
Sequence step 4: 4
Sequence step 5: 2
Sequence step 6: 1
```

Result: The program correctly runs the Collatz for the inputted number(5).

Test 4: Interval Calculation – first Interval number – Input validation

```
Enter the first interval number for Collatz:4
Enter the second interval number for Collatz:3
you must enter a number larger than 4 !
Enter the second interval number for Collatz:b
you must enter a number larger than 4 !
Enter the second interval number for Collatz:*
you must enter a number larger than 4 !
Enter the second interval number for Collatz:4
you must enter a number larger than 4 !
Enter the second interval number for Collatz:10
```

Result: The program properly handles input validation for each circumstance.

Test 5: Interval Calculation – second Interval Number – Input validation

```
Enter the first interval number for Collatz:4
Enter the second interval number for Collatz:-5
you must enter a number larger than 4 !
Enter the second interval number for Collatz:0
you must enter a number larger than 4 !
Enter the second interval number for Collatz:a
you must enter a number larger than 4 !
Enter the second interval number for Collatz:4
you must enter a number larger than 4 !
Enter the second interval number for Collatz:&
you must enter a number larger than 4 !
Enter the second interval number for Collatz:10
```

Result: The program properly handles input validation for each invalid number condition, and successfully runs upon the input of a valid number.

Test 6: Interval result output and statistics

```
Enter the second interval:
Starting number: 4
Sequence step 1: 4
Sequence step 2: 2
Sequence step 3: 1

Starting number: 5
Sequence step 1: 5
Sequence step 2: 16
Sequence step 3: 8
Sequence step 4: 4
Sequence step 5: 2
Sequence step 6: 1

Starting number: 6
Sequence step 1: 6
Sequence step 2: 3
Sequence step 3: 10
Sequence step 4: 5
Sequence step 5: 16
Sequence step 6: 8
Sequence step 7: 4
Sequence step 8: 2
Sequence step 9: 1

Starting number: 7
Sequence step 1: 7
Sequence step 2: 22
Sequence step 3: 11
Sequence step 4: 34
Sequence step 5: 17
Sequence step 6: 52
Sequence step 7: 26
Sequence step 8: 13
Sequence step 9: 40
Sequence step 10: 20
Sequence step 11: 10
Sequence step 12: 5
Sequence step 13: 16
Sequence step 14: 8
Sequence step 15: 4
Sequence step 16: 2
Sequence step 17: 1

Starting number: 8
Sequence step 1: 8
Sequence step 2: 4
Sequence step 3: 2
Sequence step 4: 1
```

```
Starting number: 8
Sequence step 1: 8
Sequence step 2: 4
Sequence step 3: 2
Sequence step 4: 1

Starting number: 9
Sequence step 1: 9
Sequence step 2: 28
Sequence step 3: 14
Sequence step 4: 7
Sequence step 5: 22
Sequence step 6: 11
Sequence step 7: 34
Sequence step 8: 17
Sequence step 9: 52
Sequence step 10: 26
Sequence step 11: 13
Sequence step 12: 40
Sequence step 13: 20
Sequence step 14: 10
Sequence step 15: 5
Sequence step 16: 16
Sequence step 17: 8
Sequence step 18: 4
Sequence step 19: 2
Sequence step 20: 1

Starting number: 10
Sequence step 1: 10
Sequence step 2: 5
Sequence step 3: 16
Sequence step 4: 8
Sequence step 5: 4
Sequence step 6: 2
Sequence step 7: 1
```

```
The number with Maximum Iterations:
Number:: 9
The maximum count of iterations:: 19

The number with the Least Iterations:
Number:: 4
iterationMinCount: 2

The number with the highest value:
Number:: 7
The highest value is:: 52
```

Results: As demonstrated from the sequence numbers 4 through 10, the calculated result returned was accurate. As well as the estimated statistics being returned with valid results.

Test 7: Input validation for program continuation/termination

```
-----  
Do you wish to continue? Y/N: 5  
Y FOR YES. OR N FOR NO.  
-----  
Do you wish to continue? Y/N: 0  
Y FOR YES. OR N FOR NO.  
-----  
Do you wish to continue? Y/N: -9  
Y FOR YES. OR N FOR NO.  
-----  
Do you wish to continue? Y/N: babc  
Y FOR YES. OR N FOR NO.  
-----  
Do you wish to continue? Y/N: yna  
Y FOR YES. OR N FOR NO.  
-----  
Do you wish to continue? Y/N: *  
Y FOR YES. OR N FOR NO.  
-----  
Do you wish to continue? Y/N: yn  
Y FOR YES. OR N FOR NO.  
-----
```

Result: The program successfully prevents any result besides the provided assumptions to continue or terminate the program

Test 8: Program continuation and termination

Case A: The user wishes to continue

Case A.1: The user inputs “Y”:

```
-----  
Do you wish to continue? Y/N: Y  
--Welcome, here is a practical demonstration of the Collatz sequence!--
```

Case A.2: The user inputs “y”:

```
-----  
Do you wish to continue? Y/N: y  
--Welcome, here is a practical demonstration of the Collatz sequence!--
```

Case A Result: In both instances the program successfully restarts.

Case B: The user wishes to terminate:

Case B.1: The user inputs “N”:

```
-----  
Do you wish to continue? Y/N: N  
Alright. Geros Dienos.
```

Case B.2: The user inputs “n”:

```
-----  
Do you wish to continue? Y/N: n  
Alright. Geros Dienos.
```

Case B Result: In both instances the program successfully terminates.

Conclusion

The tests confirmed that the program functions as intended in the listed circumstances. All required features were successfully implemented (append constructor to add start number, separate interval and statistics method). Input validation correctly handled potentially invalid user input, thus preventing errors, while the program continuation and termination behavior behaved as expected. Based on these results, the program returns accurate Collatz sequences as well as valid statistical results for user-provided intervals.

References:

Microsoft. (n.d.). Visual Studio Code [Computer Software]. From <https://code.visualstudio.com>